

Aegis Perception

Transformer Architecture & Experiment Proposal for Adverse-Weather Road-Object Detection

Professor Meeting Brief | Revised 27 August 2026

Actual project objective	Build a road-object detection pipeline that remains accurate under smoke, rain, fog and snow, and deploy the final end-to-end pipeline on an NVIDIA Jetson Nano.
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Core research question

Which transformer-based restoration architecture gives the best detection-accuracy versus latency/memory trade-off under multiple adverse-weather conditions, and can it run usefully on Jetson Nano?

Component	Role in the project	Research status
Weather restoration	Transformer-based front end removes or suppresses smoke/rain/fog/snow while preserving road-object evidence.	Primary architecture research.
Object detection	Lightweight detector predicts vehicles, pedestrians, motorcycles, signs/lights or the approved class set.	Primary task outcome.
Jetson Nano	Final end-to-end target: camera/input → restoration → detection.	Hard memory/latency constraint from the start.
Restore/bypass classifier	Potentially avoids unnecessary restoration.	Optional later work; excluded from the core proposal.

Recommended direction	Use TransWeather as the direct multi-weather transformer baseline; compare Restormer and a lightweight RAMiT-style model; then develop a compact weather-conditioned hybrid transformer optimized jointly for restoration, detection utility and Jetson constraints.
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The architecture must be selected by experiments—not by PSNR or parameter count alone.

1. What the last five years of research actually imply

The important conclusions that change our architecture and experiment design

Evidence from research	Implication for Aegis Perception
TransWeather (CVPR 2022) uses one transformer encoder-decoder with learned weather queries for rain/fog/snow.	A unified weather-conditioned transformer is a valid starting point and direct baseline.
Restormer (CVPR 2022) replaces expensive spatial attention with efficient channel-oriented attention for high-resolution restoration.	Do not use vanilla global self-attention at road-image resolution; use efficient attention inside a multi-scale hierarchy.
PromptIR (NeurIPS 2023) and weather-general/specific learning (CVPR 2023) show the value of degradation prompts and separating shared from weather-specific features.	Use a shared backbone with small condition-specific prompts/adapters instead of four completely separate large models.
RAMiT (CVPRW 2024) mixes spatial/channel attention in a lightweight hierarchy; SFHformer (ECCV 2024) combines local spatial and global frequency processing.	Candidate lightweight building blocks should be profiled on Nano before final architecture selection.
Continuous adverse-weather removal via degradation-aware distillation (CVPR 2025) addresses all-in-one model capacity and awareness.	Teacher-student distillation is a strong deployment route after a larger transformer establishes an accuracy ceiling.
SSGformer (ICCV 2025) and newer methods use spectral/spatial structure for diverse weather.	Frequency/edge priors are a focused ablation because object boundaries matter for detection.
Robust Object Detection in Challenging Weather (WACV 2024) found real all-weather detector training stronger than its denoise-then-detect strategy.	Restoration is not automatically useful. Compare against an adverse-weather-trained detector and optimize for task utility.
RT-DETR (CVPR 2024) makes DETR more efficient, but its reported real-time results use much stronger hardware than Nano.	Use a nano-scale YOLO detector for deployment and RT-DETR only as an optional non-edge reference unless Nano profiling proves otherwise.

Central research gap	Existing papers usually optimize restoration quality, detection robustness, or efficiency separately. Our defensible contribution is a compact multi-weather transformer selected and trained for downstream road-object detection, with a complete Jetson Nano accuracy-latency-memory evaluation.
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What not to claim

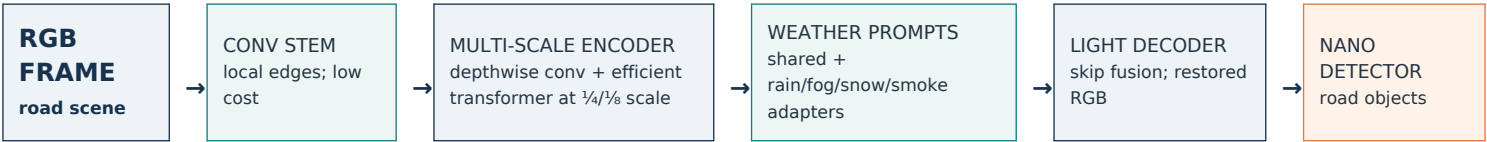
- Attention gates in the old U-Net are not new enough by themselves; they remain a historical CNN baseline.
- A visually cleaner result or higher PSNR does not establish safer ADAS perception.
- Desktop-GPU FPS does not establish Jetson Nano feasibility.
- A model trained separately for each dataset cannot automatically be called a general multi-weather solution.

2. Architecture candidates and recommended model

First reproduce known models; then introduce only a justified compact architecture

Candidate	Why include it	Expected issue	Use
Attention-Pix2Pix	Existing semester-5 direction; cheap CNN reference.	Weak novelty; 256×256 bottleneck.	Historical baseline.
TransWeather	Direct all-weather transformer with weather queries.	Must measure memory, ONNX support and real detector benefit.	Primary literature baseline.
Restormer / reduced-width Restormer	Strong efficient high-resolution restoration blocks.	Original scale may be too heavy for Nano and is not inherently all-weather-conditioned.	Accuracy ceiling + tiny variant.
PromptIR-small	Degradation prompting for all-in-one blind restoration.	Restormer-based versions may exceed edge budget.	Prompting ablation/reference.
RAMiT-style lightweight hierarchy	Designed for lightweight restoration with local/global attention.	Requires adaptation and validation for fog/snow/smoke.	Edge-oriented building block.

Recommended proposed architecture: LiteWeatherFormer



Design choice	Reason
Convolutions at full resolution	Cheaply preserve edges, texture and small objects; avoid quadratic attention on dense pixels.
Efficient transformer blocks only after downsampling	Capture global haze/smoke and long rain structures within Nano compute limits.
Shared backbone + tiny weather prompts/adapters	One deployable model while retaining weather-specific behavior.
Skip connections with controlled fusion	Preserve object boundaries; ablate old attention gates rather than assuming they help.
Detector-aware fine-tuning	Penalize restoration that removes or distorts features required for road-object detection.

Training strategy	Stage 1: paired restoration pretraining with reconstruction/structural losses. Stage 2: freeze or partially freeze the detector and fine-tune restoration with detector feature/task loss. Stage 3: distil and quantize the best model for Nano.
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3. Data and evaluation protocol

Paired datasets train restoration; labelled adverse road scenes establish detection value

Purpose	Data	What it supports	Constraint
Smoke baseline	Recover exact semester-5 paired smoke data	Old CNN reproduction and smoke restoration	Must recover provenance, split and any checkpoint.
Rain restoration/detection	RainCityscapes or another labelled paired driving set	Clean/adverse/restored same-scene analysis	Freeze scenes, not random image variants.
Fog	Foggy Cityscapes; SynFog optional external set	Paired restoration and road-scene detection	Separate synthetic validation from real transfer claims.
Snow	Snow100K/CSD subject to road labels	Restoration; detection only where labels exist	Optional fourth weather if schedule is constrained.
Real adverse test	DAWN, reported by weather and object class	Real-world detection transfer	No matched clean DAWN; use raw/resize/restored only.
All-weather detector	BDD100K/approved labelled driving set	Clean-trained vs adverse-trained detector baseline	Class mapping must be frozen across every experiment.

Controlled comparison for every labelled scene

Condition	Why it is required
Raw adverse image	Measures the actual detector problem and the no-restoration baseline.
Resize-control image	Uses the restorer's resize/downsample path without learned restoration; isolates resolution damage.
CNN-restored image	Historical Attention-Pix2Pix/vanilla baseline.
Transformer-restored image	TransWeather, Restormer-tiny/RAMiT-style and proposed LiteWeatherFormer.
Clean image where truly paired	Upper reference for paired/synthetic sets only; never fabricate a clean DAWN comparison.

Metric hierarchy

Priority	Metrics
1 — ADAS task	mAP@50 and mAP@50:95; precision/recall; per weather, severity, class and object size; missed pedestrians/vehicles and false positives.
2 — edge system	End-to-end batch-1 p50/p95 latency, FPS, peak memory, average power, joules/frame and thermal stability.
3 — restoration	PSNR/SSIM on held-out paired data; LPIPS if reproducibly available; visual failure cases.
4 — model complexity	Parameters, MACs/FLOPs at each tested resolution, ONNX/TensorRT operator compatibility and model size.

Jetson reality

Jetson Nano has a 128-core Maxwell GPU and 4 GB LPDDR4. Therefore memory, operator support and measured end-to-end latency—not a desktop FLOP estimate—must eliminate infeasible architectures early.

4. Experiment programme

Experiments answer architecture, detection, generalization and deployment questions in that order

ID	Experiment	Comparison / ablation	Decision produced
E0	Data and artifact audit	Licenses, hashes, counts, corruption, class maps, scene leakage; recover old smoke artifacts.	Whether training evidence will be valid.
E1	Historical reproduction	Vanilla Pix2Pix vs Attention-Pix2Pix vs L1-only; seeds 42/43/44.	Trustworthy CNN baseline.
E2	Architecture feasibility screen	TransWeather, Restormer-tiny, PromptIR-small/RAMiT-style at 256/384/512; parameters, MACs, GPU latency/memory and ONNX export.	Two transformer candidates allowed into full training.
E3	Transformer accuracy benchmark	Selected transformers vs CNN baseline on smoke/rain/fog; same splits and training budget.	Restoration-quality and compute Pareto front.
E4	All-in-one design	Separate per-weather models vs shared transformer vs shared + weather prompts/adapters.	Whether one deployable model handles all weather.
E5	Transformer-block ablations	Full-resolution vs low-resolution attention; channel vs window/mixed attention; prompts on/off; spectral/edge branch on/off.	Which components justify the proposed architecture.
E6	Loss ablations	L1; L1+SSIM; +edge; +frozen-detector feature/task loss.	Whether task-aware training preserves object evidence.
E7	Central detection experiment	Raw vs resize-control vs CNN-restored vs transformer-restored with the same detector.	Does restoration improve detection at all?
E8	Detector strategy	Clean-trained nano detector vs adverse-trained detector vs adverse-trained + restoration; optional RT-DETR reference off-device.	Whether restoration beats simpler detector training.
E9	Real-weather generalization	DAWN raw/resize/restored by weather/class; no matched-clean claim.	Whether gains transfer beyond paired datasets.
E10	Compression	Width/depth scaling, distillation, structured pruning, FP16 and calibrated INT8.	Smallest model retaining acceptable mAP.
E11	Jetson Nano deployment	Restorer, detector and end-to-end pipeline; sustained run under fixed power mode.	Final accuracy-latency-memory-power result.

Mandatory stop rule

If adverse-weather detector training matches or beats restoration, or if the restoration gain disappears against the resize control, do not claim that restoration improves ADAS. Reframe toward the lighter robust detector or task-aware feature enhancement.

5. Novelty, publication story and Jetson criteria

A coherent method contribution—not a collection of unrelated modules

Contribution	Specific claim to test	Required evidence
Lightweight hybrid transformer	High-resolution convolution + low-resolution efficient attention gives a better restoration/detection/compute trade-off than existing baselines.	E2, E3 and E5 across multiple resolutions/weather types.
Weather-conditioned shared model	Small prompts/adapters reduce task interference while retaining one deployable model.	E4 against separate models and a plain shared model.
Detector-aware restoration	Task/feature loss preserves small road objects better than perceptual training alone.	E6/E7 with per-size and per-class detection, including harmful cases.
Edge-constrained evidence	The contribution remains useful after distillation/quantization on Jetson Nano.	E10/E11 with accuracy, sustained latency, memory and power.

Strong paper statement

A detector-aware, weather-conditioned lightweight restoration transformer for road-object perception, evaluated from paired restoration quality through real adverse-weather detection to sustained Jetson Nano deployment.

Provisional deployment acceptance criteria — professor to approve

Criterion	Provisional target / reporting rule
Fit	Restorer + detector + runtime fit within 4 GB without swap; report peak resident/device memory.
Throughput	Target ≥ 10 end-to-end FPS as an initial engineering goal; always report p50/p95 and sustained performance, not only a best frame.
Accuracy	Restoration should improve adverse-weather mAP beyond raw and resize-control; report weather/class/object-size breakdown.
Quantization	Report FP32/FP16/INT8 accuracy drift separately for restorer, detector and full pipeline.
Power/thermal	Fix Nano power mode, record temperature/power and run long enough to expose throttling.

Safe and ambitious publication paths

Path	Minimum result
Workshop / applied venue	Rigorous E0–E4 and E7–E11 study, honest negative results, real DAWN transfer and complete Nano profiling.
Stronger conference submission	Above evidence plus a clearly better LiteWeatherFormer Pareto front, prompt/task-loss ablations, multi-weather generalization and statistical confidence.

6. Immediate plan and decisions required

The next meeting should freeze choices needed to start E0-E2

Phase	Work	Deliverable
Week 1	Recover smoke data/artifacts; finalize classes and datasets; establish GPU/Nano environments.	Dataset cards, fixed manifests and hardware/software inventory.
Weeks 2-3	Reproduce CNN baseline; obtain/run TransWeather, Restormer and RAMiT/PromptIR candidates on a small common subset.	E1 results + E2 feasibility table.
Weeks 4-6	Train two selected transformers; run all-in-one/prompt and block/loss ablations.	E3-E6 tables; freeze proposed model.
Weeks 7-8	Detection matrix and adverse-trained detector comparator; DAWN transfer.	E7-E9 central evidence.
Weeks 9-10	Distil/scale/quantize; deploy to Nano; sustained benchmark.	E10-E11 Pareto and edge tables.
Final 2 weeks	Freeze results, figures, failure analysis and paper.	Submission-ready manuscript and reproducibility package.

Professor decisions

- Confirm that the transformer is primarily the restoration/front-end architecture, while the deployable detector remains lightweight.
- Approve smoke + rain + fog as mandatory and snow as schedule-dependent, or specify another minimum set.
- Approve TransWeather, Restormer-tiny and RAMiT/PromptIR-small for the E2 candidate screen.
- Freeze the object classes, detector family/version and datasets before training.
- Confirm GPU allocation, Jetson Nano/JetPack version, storage and the provisional ≥ 10 FPS target.
- Approve LiteWeatherFormer—hybrid low-resolution attention, weather prompts and detector-aware loss—as the novelty hypothesis after E2.

Key primary papers

1. TransWeather — CVPR 2022. [Primary link](#)
2. Restormer — CVPR 2022. [Primary link](#)
3. Weather-General and Weather-Specific Features — CVPR 2023. [Primary link](#)
4. PromptIR — NeurIPS 2023. [Primary link](#)
5. Robust Object Detection in Challenging Weather — WACV 2024. [Primary link](#)
6. RAMiT — CVPR Workshops 2024. [Primary link](#)
7. SFHformer — ECCV 2024. [Primary link](#)
8. RT-DETR — CVPR 2024. [Primary link](#)
9. Continuous Adverse Weather Removal via Degradation-Aware Distillation — CVPR 2025. [Primary link](#)
10. SSGformer — ICCV 2025. [Primary link](#)
11. NVIDIA Jetson Nano specifications. [Primary link](#)

Required meeting outcome

Freeze the architecture-screen candidates, datasets/classes, detector, resource allocation and Nano acceptance criteria. Then begin data audit, CNN reproduction and transformer feasibility profiling immediately.